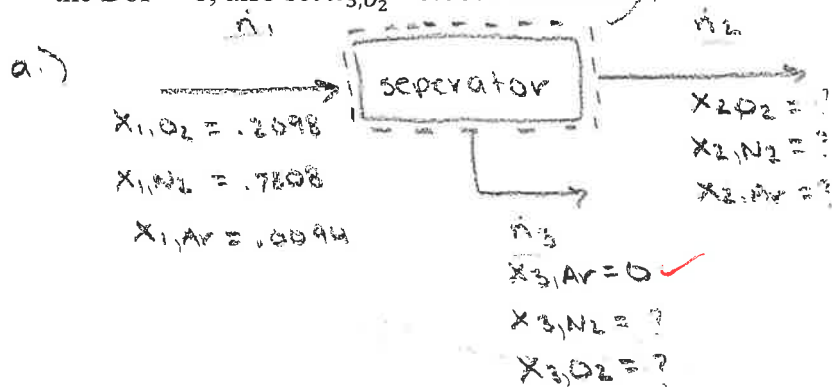


To Receive Full Credit, You Must Show All Your Work

The Linde-Frankl separation process uses a feed of atmospheric air (Stream 1) which has a composition of 20.98 mol% oxygen (O_2), 78.08 mol% nitrogen (N_2), and 0.94 mol% argon (Ar) to produce a product (Stream 2) enriched in oxygen and a byproduct (Stream 3) which is enriched in nitrogen and contains no argon.

- (a) Draw and label a block flow diagram. Define a system. For the 3 streams, use $\dot{n}_i$ for total molar flow rates and x_{i,O_2} , x_{i,N_2} , and $x_{i,Ar}$ for mole fractions for oxygen, nitrogen, and argon, respectively. Use $i = 1, 2, \& 3$.
- (b) Do a DoF analysis using p. 105 of Murphy. Give values for 4a, 4b, 5a, 5b, 5c, and 5d. Compute the DoF.
- (c) If the DoF < 0, stop here. The problem is overspecified. If DoF = 0, solve the problem. If DoF = 1, set $\dot{n}_2 = 100$ kmol/hr and solve. If DoF = 2, also set $x_{2,O_2} = 0.9$ and solve. If the DoF = 3, also set $x_{3,N_2} = 0.953$ and solve. If the DoF = 4, also set $\dot{n}_1 = 224$ kmol/hr and solve. If the DoF = 5, also set $x_{2,Ar} = 0.091$ and solve. If the DoF = 6, also set $x_{3,O_2} = 0.067$ and solve.



b.) 4a - 8 4b - 0
5a - 0 5b - 2 5c - 0
5d - 3

DoF: $8 - (3 + 2) = 3$

c.) since DoF = 3, also given $\dot{n}_2 = 100 \frac{\text{kmol}}{\text{hr}}$ and $x_{2,O_2} = 0.9$ and

$\dot{n}_1 = \dot{n}_2 + \dot{n}_3$ ①

$O_2: (0.2098)\dot{n}_1 = (0.9)(100 \frac{\text{kmol}}{\text{hr}}) + x_{3,O_2}(\dot{n}_3)$ ②

$N_2: (0.7808)\dot{n}_1 = x_{2,N_2}(100) + x_{3,N_2}\dot{n}_3$ ④

$Ar: (0.0094)\dot{n}_1 = x_{2,Ar}(100 \frac{\text{kmol}}{\text{hr}})$ ③

$(0.0094)(100 + \dot{n}_3) = x_{2,Ar}(100) \rightarrow 0.94 + 0.0094\dot{n}_3 = 100 x_{2,Ar}$

	$\dot{n}_1$	$\dot{n}_3$	x_{2,N_2}	$x_{2,Ar}$	x_{3,O_2}
1	✓	✓			
2	✓	✓			✓
3	✓			✓	
4	✓		✓		

go to back

$$N_2) 7808 \dot{n}_1 = 100 x_{2,N_2} + .953 \dot{n}_3$$

$$\dot{n}_1 = 100 + \dot{n}_3$$

$$.7808(100 + \dot{n}_3) = 100 x_{2,N_2} + .953 \dot{n}_3$$

$$x_{3,O_2} = 1 - x_{3,N_2}$$

$$x_{3,O_2} = 1 - .953 = .047$$

$$O_2) .2098 \dot{n}_1 = (.9 \cdot 100) + (.047) \dot{n}_3$$

$$.2098(100 + \dot{n}_3) = 90 + .047 \dot{n}_3$$

$$.1628 \dot{n}_3 = 69.02$$

$$\dot{n}_3 = 423.956 + 100$$

$$\dot{n}_3 = 423.956 \frac{\text{kmol}}{\text{hr}}$$

$$x_{3,O_2} = .047$$

$$x_{3,N_2} = .953$$

$$\dot{n}_1 = 523.956 \frac{\text{kmol}}{\text{hr}}$$

$$.7808(523.956) = 100 x_{2,N_2} + .953(423.956)$$

$$x_{2,N_2} = .0507$$

Ar)

$$.94 + .0094(423.956) = 100 x_{2,Ar}$$

$$x_{2,Ar} = .6493$$

great work!